

BSCS FINAL PROJECT · UNIVERSITY OF CENTRAL PUNJAB

PLAYERSZWZ

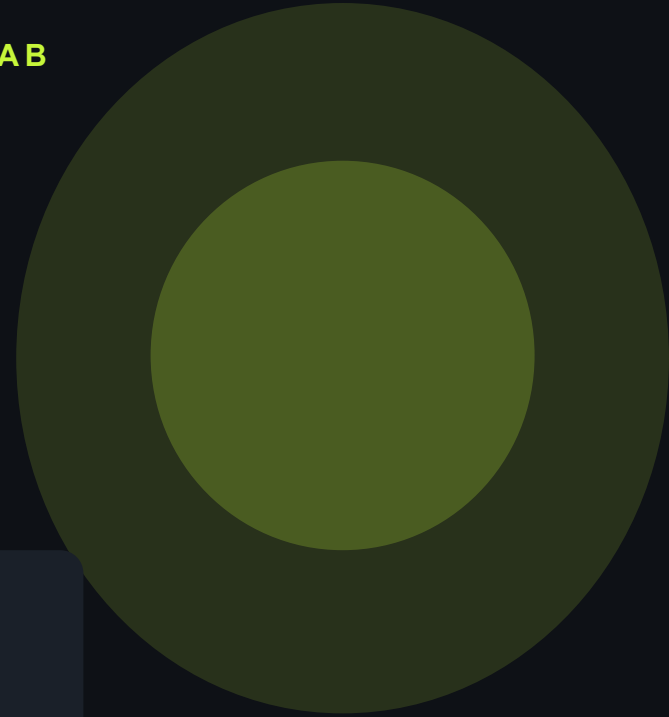
AI-Powered Sports Matchmaking & Arena Booking
Application

Group ID **S26CS024**

TEAM

Waleed Abid · Afaq Ul Islam · Hassan Ahmed

Faculty of Information Technology & Computer Science



Team



Waleed Abid

L1F24PACS0021

Mobile app architecture,
backend integration and
project coordination.



Afaq UI Islam

L1F21BSCS0951

Scrum Master, frontend
implementation, maps /
location services and UI flows.



Hassan Ahmed

L1F22BSCS1053

AI recommendation module,
testing, documentation and
QA.

Introduction & Background

PLAYERSZWZ streamlines organizing casual sports matches — discovering players, creating and joining games, booking arenas, and building trust — all inside one mobile workflow.



Problem Context

- Informal matches get delayed or cancelled
- Incomplete teams & late confirmations
- Ground booking scattered across calls & WhatsApp



Product Vision

- A single, structured mobile platform
- Reduce coordination friction end-to-end
- Reliability scores to trust unknown players



Key Stakeholders

- Players & Game Organizers
- Arena Managers & Administrator
- Product Owner & Development Team

Agile Framework Setup

PROJECT EPICS

EP-01

User Account
Management

EP-02

Player Profile &
Preferences

EP-03

Nearby Discovery

EP-04

Game Management

EP-05

Arena Discovery &
Booking

EP-06

AI Recommendation
Engine

EP-07

Feedback, Rating &
Moderation

EP-08

Notification System

SCRUM ROLES

Scrum Master

Afaq UI Islam

Development Team

Waleed · Afaq · Hassan

SPRINT LENGTH · 2 WEEKS

TOOLS

Trello / Jira · Git & GitHub · Figma ·
Firebase Console

Sprint Planning & Sprint Backlog

9

Sprint 1 US-01 → US-03

Project setup, authentication, base DB models & profile screens

14

Sprint 2 US-04 → US-07

Player discovery, game creation, join requests & organizer approval

15

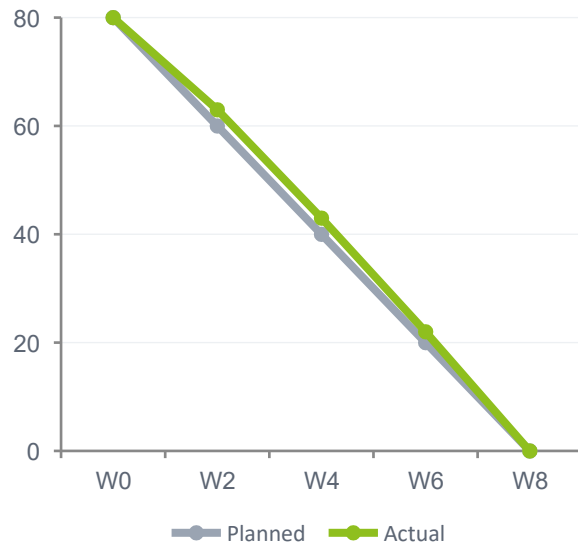
Sprint 3 US-08, 09, 13

Arena listing, booking workflow, notifications & status updates

19

Sprint 4 US-10 → 14

Recommendation logic, rating / report, admin moderation, testing & docs

SPRINT BURNDOWN

Functional Requirements

MODELED THROUGH



High-Level Architecture



Use-Case Diagram



Class Diagram



Sequence Diagrams



ER Model & Schema

ACTORS & CORE USE CASES

Player

Register / Login · Manage Profile · Discover Players · Join Game · Rate / Report · Recommendations

Game Organizer

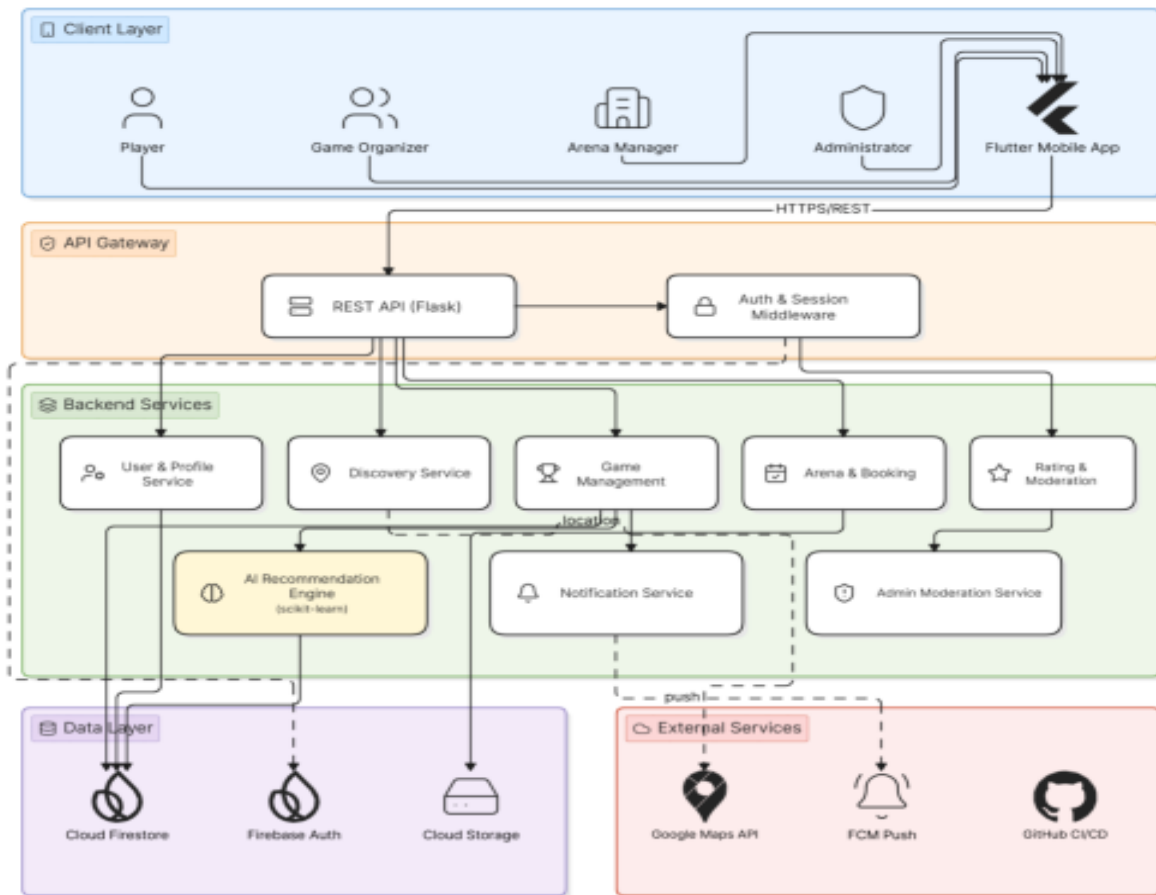
Create Game · Approve / Reject Requests · Arena Discovery & Booking

Administrator

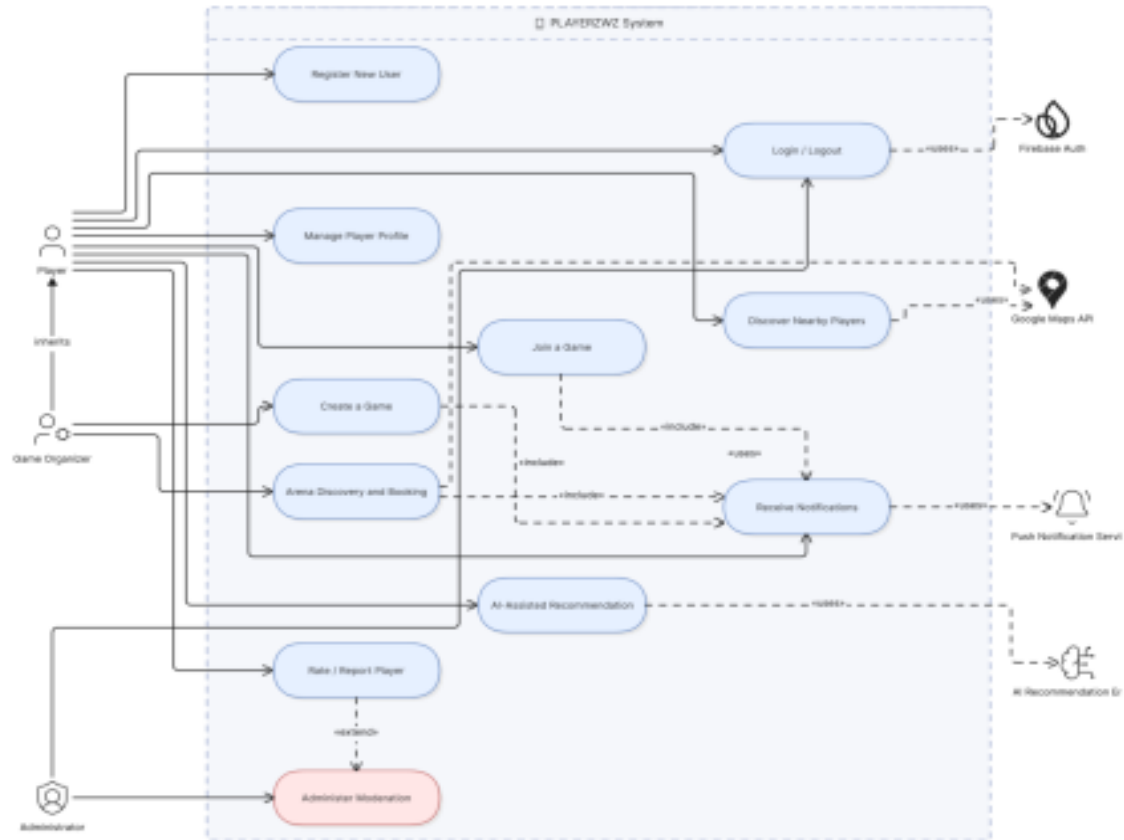
Review Reports · Warn / Suspend / Clear · Audit Log

EXTERNAL Firebase Auth · Google Maps API · FCM Push · AI Recommendation Engine

Architecture Diagram



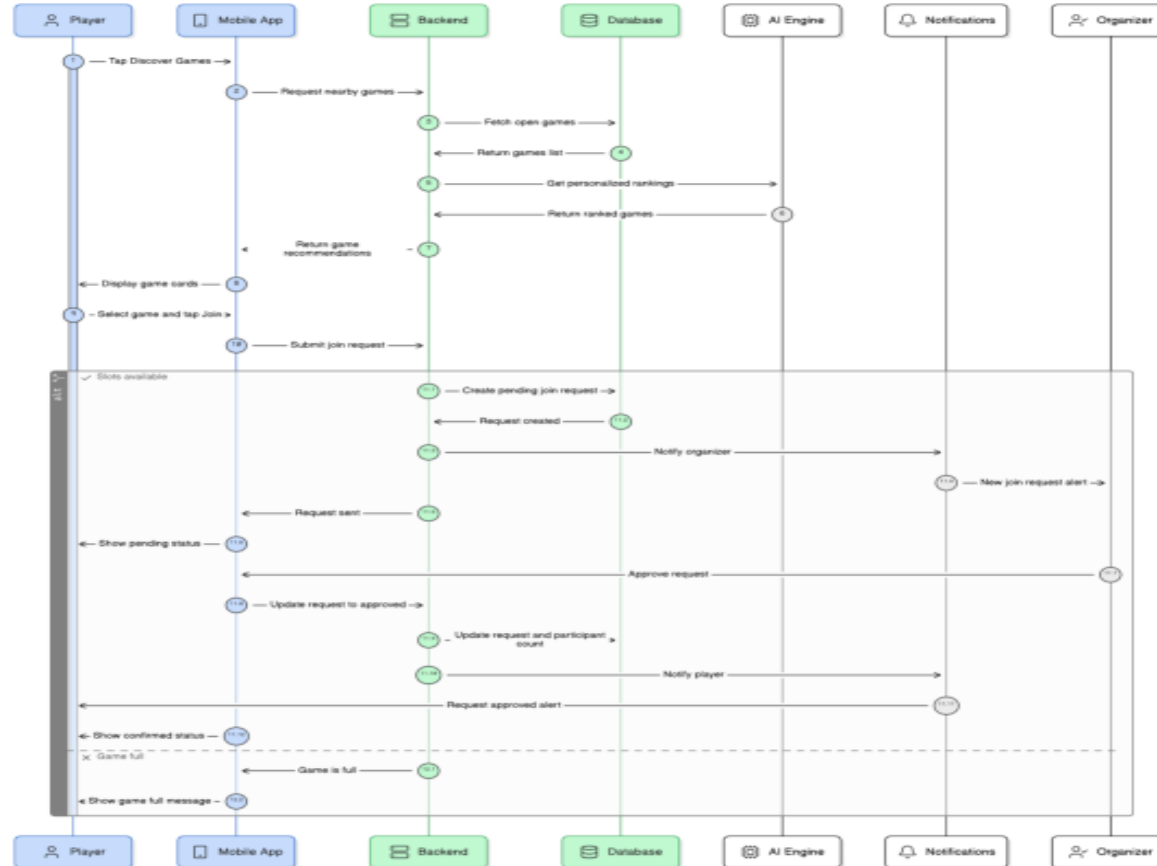
Use Case Diagram

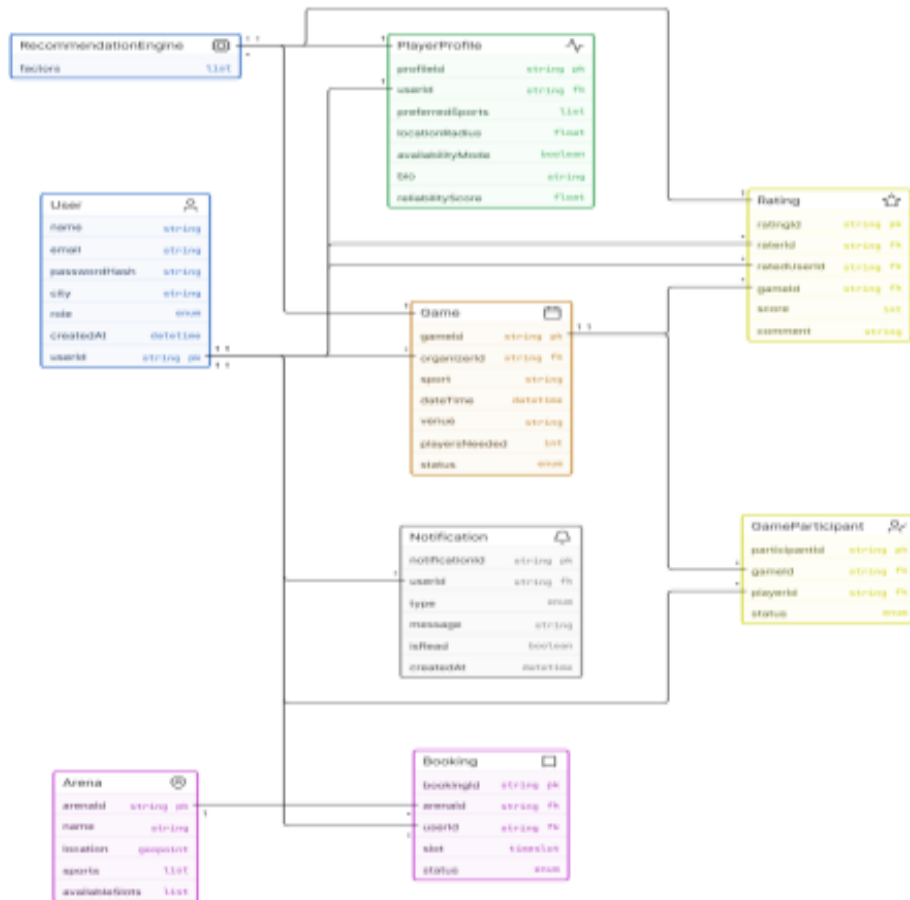


Class Diagram

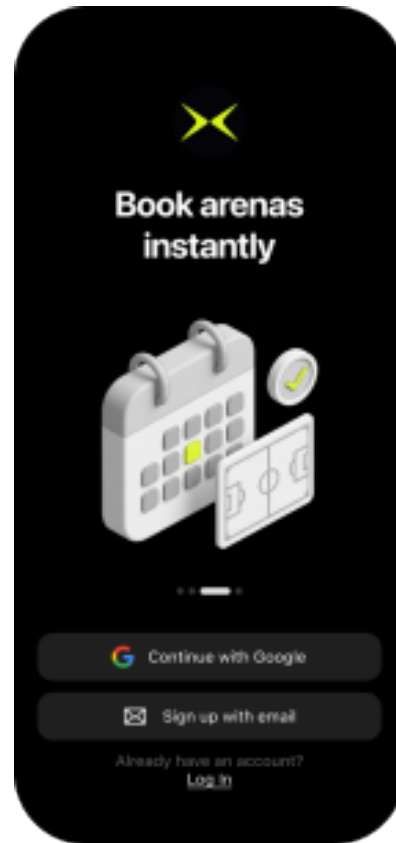
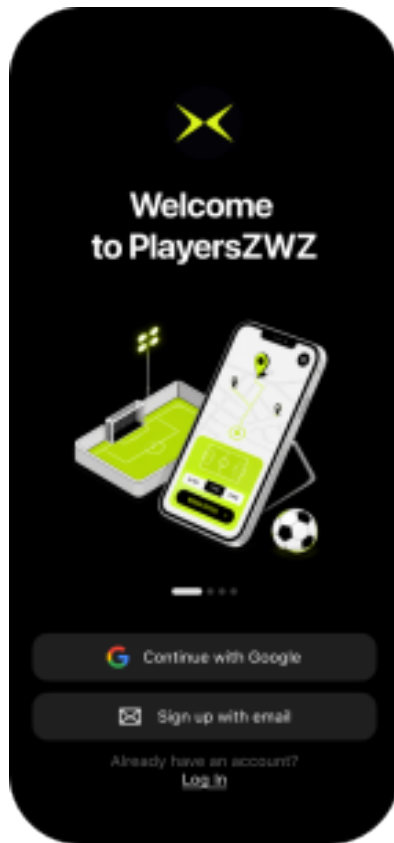
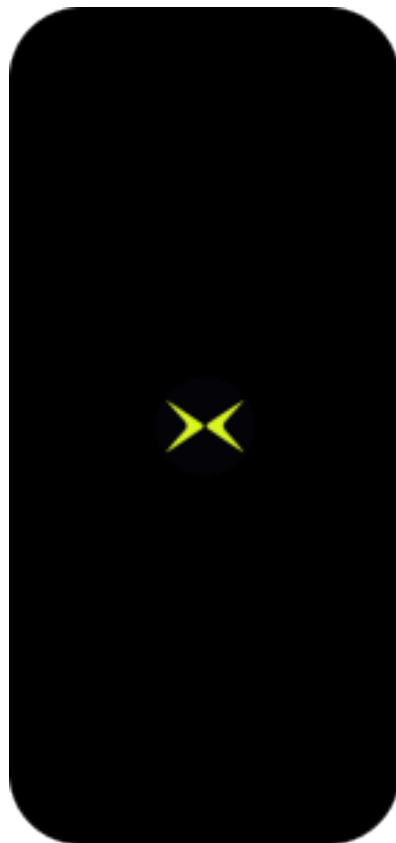


Sequence Diagram





08 · WORKING PROTOTYPES





Create your account

First name

Last name

Email address

Password

Confirm password

Create account

or

 Continue with Google

Already have an account?
[Log in](#)

Create profile



@ataq_ashir

Your username can only contain alphabets, numbers, underscores and dots.

Next

Pick your sports

Personalize your match feed.

Basketball

Badminton

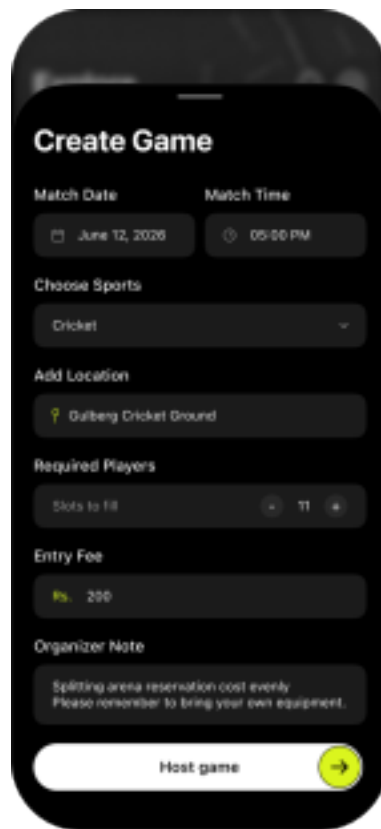
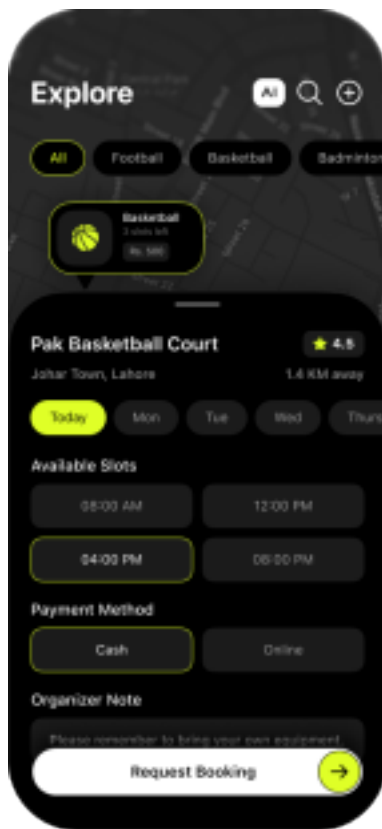
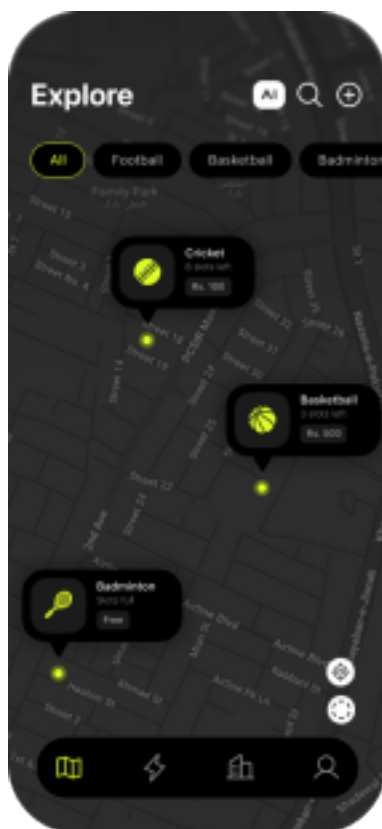
Football

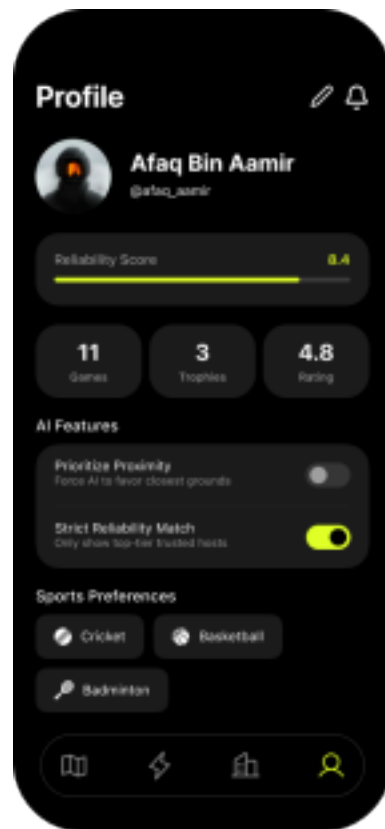
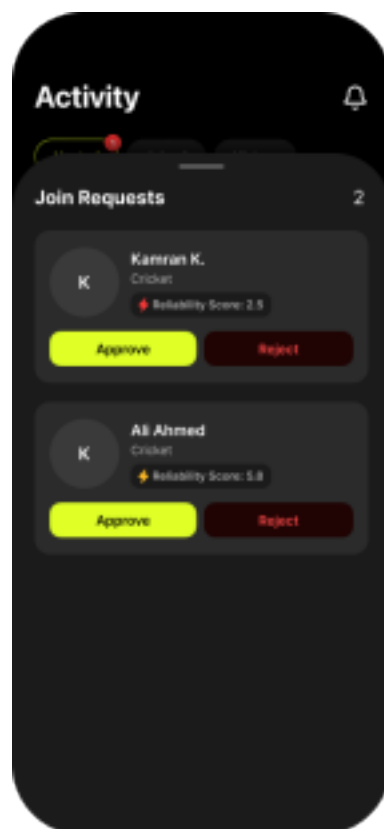
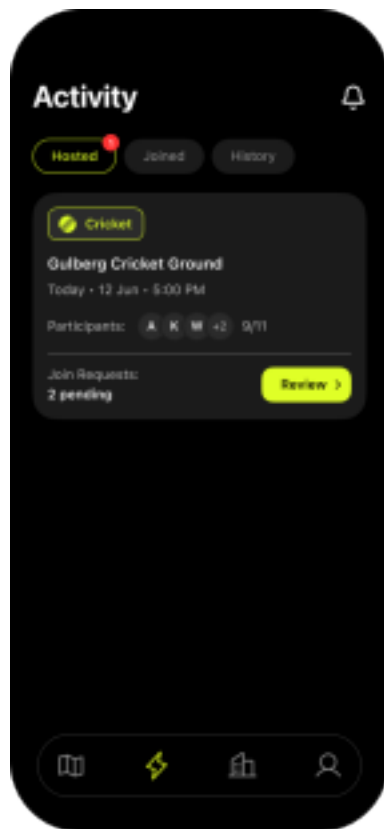
Skating

Cricket

Tennis

Continue





Non-Functional & Quality Attributes



Performance

Dashboard < 3s · nearby search
first results < 5s · rule-based rec <
3s, history-based < 6s.



Security

No plaintext passwords · Firebase
Auth + verified tokens · sensitive
actions logged.



Usability

Self-serve register / create / join ·
manual area fallback · clear filters &
error states.



Maintainability

Modular UI / API / services / data
layers · each module independently
testable.



Scalability

≥ 50 concurrent test users ·
Firestore + geohash / radius +
paginated queries.



Reliability & Fairness

Consistent join / booking states ·
reliability changes auditable & rule-
based.

Revised Plans & Risk Analysis

TOP RISKS & MITIGATION

Risk	Impact	Mitigation
Location permission denied	High	Manual area entry + explain location benefit
Weak rec. for inactive users	Med	Rule-based fallback (sport / area / availability)
Outdated arena slot data	High	Request-based bookings; confirm before Approved
Unfair ratings hurt score	High	Block double rating; dispute window; confirmed no-show only
Scope too large for Phase 1	High	Exclude payments, tournaments, intl deployment

REVISED PROJECT PLAN

- ✓ Phase-I SRS **Done**
- ✓ Phase-II SDS Architecture **Done**
- ✓ UML & Database Design **Done**
- ✓ API & UI/UX Design **Done**
- Testing & IV&V **In Progress**
- Prototype Sprint 1 & 2 **Planned**

Conclusion

8

Epics

14

User Stories

4

Sprints

57

Story Points



Progress So Far

- Phases I & II complete — requirements & full design baseline
- Architecture, UML, ERD, API contracts & UI/UX flows defined
- Working dark-themed mobile UI prototype built



Future Targets

- ML recommendations from real participation history
- Arena Manager portal + calendar slot management
- Tournaments / leagues & secure payment integration
- iOS support & web admin dashboard



Thank You

Q&A Session

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